CSCI 131 — Introduction to Programming in Python
Programming Assignment 7
Strings and Lists as Sequences

There are no starter files. Open IDLE, choose **File** → **New File** for each program, and save it under the exact name given in the problem. Put your name in a comment on the first line of each file. Where a problem gives you a list, type it into your program exactly as shown.

Problem 1 — Card Mask

Receipts and websites never show your whole card number. They show only the last four digits and replace the rest with stars. Write a program `mask.py` which asks for a card number and prints it that way. Use the prompt `'Card number:  '`. You may assume the user types only digits, with no spaces or dashes, and at least four of them.

A sample run:

```
Card number: 4111222233334444
Masked: ************4444
```

Card numbers are not all the same length: the third row has fifteen digits, so it gets one fewer star. Your program has to work out how many stars to print.

Problem 2 — Curve

Write a program `curve.py` which starts with this list of test scores:

```
scores = [72, 88, 95, 61, 84]
```

The program curves the scores so that the highest one becomes 100: it works out how many points the highest score is short of 100, and adds that many points to *every* score. It prints the size of the curve, then the curved list. For the list above, it prints:

```
Curve: 5
[77, 93, 100, 66, 89]
```

You must **change the list itself**, so that after your loop the list holds the curved scores. Then print it with a single `print(scores)` at the end of the program. A program which prints the curved scores one at a time, without changing the list, does not answer this problem.

Submission

Submit all the source files on Canvas by the end of the period. Put your name in a comment at the top of each file. If a classmate helped you, put their name in the comment as well.

If you are not finished when the period ends, submit what you have — a partial program earns partial credit, an empty one earns none. You may then complete it on your own and resubmit by Sunday, October 18 at 11:59 pm for up to 50% of the remaining credit.